

Project Milestone 2

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Customer Complaint Triage and Response

Customer complaint management is a critical function for financial institutions. There is a strong regulatory environment in place emphasizing transparency, fairness, and timely responses to customer complaints. Institutions must respond to complaints consistently with a clearly outlined process that identifies systematic issues that may indicate broader operational or compliance risks.

From a risk-mitigation perspective, complaint data is a valuable source of insight into business operations. However, traditional approaches rely heavily on manual review, rule-based categorization, and limited analytics, which do not scale well with complaint volumes.

Generative AI provides an opportunity to transform aspects of this process in several areas. The shift from rules-based systems to probabilistic, context-aware models, enabled by LLMs that can understand nuanced customer narratives, can improve complaint triage, analysis, and insights. This project explores how this can be applied to the complaint triage stage to analyze and generate responses that are consistent with requirements but still specific to the financial customer service context. This would provide a scalable, explainable, and consistently auditable systems to minimize regulatory and reputational risk.

Project Description

The proposed system will function as a pipeline to process raw customer complaints and produce structured, actionable outputs. Specifically, the system will classify complaints, summarize narratives, and draft an initial response aligned with institutional tone and regulatory expectations. The system will be designed to focus on explainability and consistency, both of which are critical in regulatory environments. The outputs must be interpretable by compliance teams and auditable. This application is particularly relevant to financial institutions where complaint mismanagement can lead to regulatory scrutiny, reputational damage, and financial penalties.

Project Goals

Primary project objectives:

- Improve operational efficiency by reducing the manual effort required to review and categorize complaints
- Enhance consistency in customer interactions by generating standardized, policy-aligned responses.
- Support risk mitigation efforts with early identification of complaint insights.
- Improve customer experience with faster, more coherent responses.

Initial approach methodology

The project will aim to use publicly available datasets containing structured and unstructured complaint narratives for analysis. In obtaining datasets, retrieving datasets with reduced personal identifiable information or actual company information will be prioritized to minimize potential ethical risks for use in the project. Data preparation will include text cleaning, tokenization, formatting, category construction, and potential input-output pairs (complaint-to-summary, category, and response). To ensure the best possible model performance, remove ambiguous or incomplete records and ensure class balance. Prompt engineering experimentation will be conducted to evaluate how well classification, summarization, and response generation are achieved at a base level and with additional structured prompts such as instructions, roles, and output structure.

The solution will leverage the OpenAI API and fine-tune a large language model (LLM). The plan is to use both a general-purpose GPT-class model and an adapted version trained on complaint-response pairs. This approach is appropriate because the task involves multiple NLP subtasks that are language-driven, including text classification for complaint categorization, summarization, context-aware response generation, and root cause analysis. Traditional models would require separate pipelines for each task. Fine-tuning is required as prompting alone could introduce variability in output structure and tone. Fine-tuning allows the model to create consistent institutional response patterns, complaint taxonomies, and risk-aware language constraints.

Evaluation will need to be conducted using both quantitative and qualitative methods, such as accuracy and human evaluation of aspects of tone and response relevance. The final goal will be an application that accepts a complaint text and generates a category, summary, and suggested response. This is an example of how a larger system could be deployed within a financial institution.

Several challenges are expected, including data quality, potential model hallucination, and primarily evaluation complexity. Assessing the quality of generated text requires subjective evaluation alongside quantitative metrics.

Prompt Experimentation

To assess the potential of this approach to achieve the project goal, five prompt experiments were conducted (see **the Experiments** section). As a result, all approaches correctly identified the example complaint type as a Billing Issue as the dominant category, even with an identification of secondary customer service dissatisfaction. There was minimal difference between zero-shot and few-shot experiments, which is likely due to the clarity of the complaint. Instruction prompting

resulted in concise, operational, and useful summaries. Both role-prompting and structured outputs were successful in producing outputs that would be valuable for use in a complaint handling pipeline.

Fine Tuning Approach

Based on the results of the prompt experimentation, the fine-tuning approach will focus on supervised training using curated data. The training data will be structured as raw customer complaint text. The model will then be trained on input-output pairs with a clear instruction format specifying output categories, summaries, and responses. This training will be monitored to identify performance and output consistency. This will be done using classification metrics, including accuracy, precision, recall, and F1 Score, as well as evaluating response quality for compliance and tone through human review.

Conclusion

This project aims to develop a generative AI-powered system that enhances the efficiency and consistency of complaint handling in financial institutions. By leveraging a fine-tuned language model, the system would transform unstructured data into actionable insights and standardized responses. This would represent a significant advancement over traditional methods and aligns with current industry trends toward AI adoption and practical applications.

The refined approach confirms that a generative AI approach is both feasible and appropriate for automation of part of the customer complaint handling process. The prompt engineering experiments demonstrate that while base LLMs can perform the required tasks, achieving the increased consistency and control required for a regulated environment should involve fine-tuning. The supervised fine-tuning will provide a strong foundation for subsequent project steps, including model development.

Experiments

Experiment 1: Zero-Shot Classification

Prompt:

“Classify the following customer complaint into one of these categories: Billing Issue, Fraud, Customer Service, Account Management.

Complaint: ‘I was charged twice for a transaction and cannot get support to fix it.’”

Output:

Category: Billing Issue

Rationale:

The core issue is a duplicate charge, which directly maps to billing, even though there is a secondary frustration with support.

Experiment 2: Few-Shot Classification with Examples

Prompt + Examples:

Example 1:

Complaint: "I see a charge on my account that I don't recognize."

Category: Fraud

Example 2:

Complaint: "Your support team never responds to my emails."

Category: Customer Service

Example 3:

Complaint: "I need help updating my contact information."

Category: Account Management

New Complaint:

"I was charged twice for a transaction and cannot get support to fix it."

Output:

Category: Billing Issue

Rationale:

Even with contextual examples, the model prioritizes the financial discrepancy (duplicate charge) over the service complaint.

Experiment 3: Summarization with Instruction Prompting

Prompt:

"Summarize the following complaint in one sentence for internal review."

Output:

The customer reports being double-charged for a transaction and is unable to resolve the issue through customer support.

Experiment 4: Role-Based Response Generation

Prompt:

"Act as a compliance officer at a financial institution. Draft a professional response to the following complaint..."

Output:

Dear Customer,

Thank you for bringing this matter to our attention. We understand your concern regarding the duplicate charge and the difficulty you have experienced in obtaining support. Please be assured that we take such issues seriously and are committed to resolving them promptly.

We are currently reviewing your account activity to identify the source of the duplicate transaction. A member of our support team will contact you shortly to provide an update and assist in resolving this matter. If the charge is confirmed to be in error, appropriate corrective action, including a refund, will be initiated without delay.

We appreciate your patience and apologize for any inconvenience caused.

Sincerely,
Compliance Office

Experiment 5: Structured Output (JSON Format)**Prompt:**

"Analyze the complaint and return output in the following JSON format..."

Output:

```
{  
  "category": "Billing Issue",  
  "summary": "Customer was charged twice for a transaction and has been unable to  
resolve the issue through support.",  
  "response": "We apologize for the inconvenience caused by the duplicate charge and  
the difficulty in reaching support. We are reviewing your account and will work to resolve  
the issue promptly, including issuing a refund if necessary."  
}
```

References

OpenAI. (2026). *ChatGPT* (Mar 27 version) [Large language model].
<https://chat.openai.com/>